

5G Mobile Networked Clusters Anomaly Detection Under Distribution Shifts: A Causal Perspective

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Abstract—Along with the application of the C-band of the high-frequency spectrum, the number of 5G wireless base stations has increased significantly compared with 3G/4G networks. Therefore, how to intelligently detect anomalies in base stations is crucial for the daily operation and maintenance of 5G networks. Nowadays, 5G Mobile Network Clusters (5G-MNCs) are gradually replacing individual base stations for maintenance as a group of regional base stations with high internal correlation. However, anomaly detection in 5G-MNC remains challenging due to the complex spatio-temporal measurement data, and additionally, the distributional variations of such data's time series can also have a large impact on anomaly detection. In this work, we propose a novel architecture that integrates causal intervention with doubly residual spatio-temporal graph convolution (Causal-DRSTGC). First, a causal intervention module is introduced to eliminate the effect of confounders due to distribution shifts. Second, a doubly residual structure is adopted to output reconstruction and hidden representations respectively, which is more conducive to promoting the representation capability of 5G-MNC data. Third, after stacking multiple DRSTGC blocks, a reconstruction module and a discrimination module are designed to process data reconstruction representation and hidden representations for joint optimization. Finally, a flexible score is designed to balance the contributions of each component to improve the overall performance of anomaly detection. Extensive experiments conducted on China Mobile's real-world dataset demonstrate that the proposed model achieves remarkable performance. For Cluster1, the F1-scores for shielding and decommissioning anomalies reach 89.29% and 90.53%, respectively. Additional validation on public industrial datasets demonstrates its generalization capability, achieving an F1-score of 83.43% on PSM, confirming its effectiveness in various application scenarios.

Index Terms—Anomaly detection, causal interventions, spatio-temporal modeling, graph convolution network, 5G mobile network.

I. INTRODUCTION

WITH its high bandwidth, ultra-low latency, and extensive connection, the 5G mobile network offers vital technical support for vertical sector applications such as smart

healthcare systems, vehicle-to-everything communications, drone communication systems, and the industrial Internet of Things (IoT) [1], [2], [3], [4].

Although 5G's millimeter wave spectrum enables increased bandwidth, it also requires a significant increase in base station density. This presents deployment challenges for wireless base stations—essential components of communication infrastructure. A vital operational priority is anomaly detection in base station clusters, which is highlighted by this rapid infrastructure growth.

Traditional approaches for mobile network anomaly detection mostly use manually set fixed criteria to track Key Performance Indicators (KPIs) at specific base stations [5], [6], [7]. Despite the interpretability benefits of these expert knowledge-based approaches, they have three key drawbacks. First, for modern networks—comprising hundreds of thousands of base station nodes—creating separate threshold models for each base station requires substantial expert input and ongoing maintenance as network topologies change. This drives operating costs to unaffordable levels. Second, current methods usually use monitoring systems with just one variable, analyzing each base station's KPIs separately. This design is still vulnerable to false alarms brought on by single-node data changes and exhibits significant sensitivity to local noise. Third, the spatial correlation features inherent in base station clusters are not sufficiently captured by existing solutions. Notably, in cellular network designs, neighboring base stations achieve spatial coordination through techniques like load balancing and handover optimization. However, the intrinsic spatio-temporal coupling properties of KPI data are not captured by conventional threshold-based methods, leading to systematic discrepancies in the assessment of the overall normal condition of the network [8].

In practical operational scenarios, compared to the discrete deployment architectures of 3G/4G networks, 5G systems exhibit a denser network topology. This denser topology facilitates resource sharing and dynamic scheduling through clustered configurations rather than relying on individual or loosely coordinated base stations [9]. A 5G-MNC can be defined as a closely coordinated cluster of base stations within specific coverage areas, as illustrated in Fig.1. To ensure continuous coverage and effective load distribution across 5G base stations, strong interconnections between wireless nodes must be maintained across the service area. This architecture enables seamless handover between multiple base stations during user mobility, ensures load balancing, and leverages collaborative virtualization technologies to achieve resource sharing and dynamic scheduling. This focus on real-time

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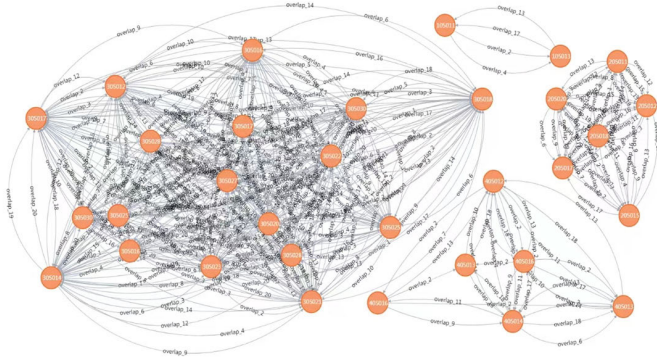


Fig. 1. Example of 5G-MNCs, and each node represents a base station.

performance aligns with advancements in related wireless communications technologies [10].

Therefore, in this study, we conduct anomaly detection in 5G mobile networks through the lens of 5G-MNC architecture. The methodological strengths are multifold: By establishing the entire 5G mobile network as a unified monitoring and modeling architecture, our approach captures spatio-temporal correlations among network nodes, consequently enhancing anomaly detection performance. System-wide anomaly detection across the 5G-MNC substantially mitigates the effects of noise and fluctuations originating from individual nodes on collective detection performance. The proposed technique effectively suppresses false positives induced by unaccounted node interactions, thereby strengthening both robustness and stability in detection processes. Moreover, this paradigm promotes data-driven learning mechanisms that remove dependency on expert-configured thresholds, achieving dual improvements in the reduction of maintenance costs and the enhancement of detection efficiency.

From another perspective, distribution shift problems with 5G-MNC data may compromise the accuracy of anomaly detection systems. Variations in the environment are the main cause of these shifts in distribution. Building environment-invariant representations of data is crucial to reducing the effects of distribution shifts. To learn invariant representations in this situation, we include a Partial Conditional Invariance Regularization (PCIR) term in this architecture [11]. By lowering the Maximum Mean Discrepancy (MMD), we decrease the disparity between representations from various environments. Therefore, due to the influence of communication environment, distribution shift problems are commonly present in intelligent wireless communication systems [12]. This research effectively eliminates the impact of distribution shifts brought on by environmental changes using causal intervention approaches.

For analyzing complex spatio-temporal interdependencies within 5G-MNC data, we develop a Doubly Residual Spatio-Temporal Graph Convolution (DRSTGC) block incorporating two graph convolution components with residual connections. The primary component systematically incorporates predefined domain knowledge, while the auxiliary component employs data-driven methodology to address residual patterns unresolved by the preceding module. This innovative architecture effectively leverages 5G-MNC's pre-

defined topological characteristics while elucidating intrinsic synchronization dependencies between base stations. Furthermore, the DRSTGC module incorporates a doubly residual architecture that simultaneously outputs reconstruction and hidden representations to enhance the representational capacity of 5G-MNC data. Following two stacked DRSTGC modules, we implement causal intervention mechanisms to mitigate distribution shift effects. Finally, we develop a dual-component architecture comprising discrimination and reconstruction modules for collaborative anomaly detection, effectively integrating information from both original data domains and latent spatial representations. The synergistic integration of these modules with an adaptive anomaly scoring mechanism enables complementary inference capabilities, thereby achieving enhanced detection robustness. The main contributions of this work are summarized as follows:

1) A novel anomaly detection model (Causal-DRSTGC) is proposed, by combining causal learning with graph learning mechanisms, which can effectively eliminate environmental confounding factors and address anomaly detection of spatio-temporal data in distribution shifts.

2) The introduction of a joint optimization mechanism enables the synergistic integration of discrimination-based and reconstruction-based modules, which are implemented in the latent and original data domains, respectively. This integration is designed to achieve superior model performance.

3) Extensive experiments on real-world 5G-MNC datasets and public industrial datasets validate the superiority of the proposed method in various application scenarios.

II. RELATED WORK

A. Mobile Network Anomaly Detection

The domain of mobile network anomaly detection has seen extensive methodological development, with various approaches investigating distinct strategies and frameworks for identifying network irregularities. Dridi et al. [13] developed an automated framework that employs machine learning techniques, including One-Class Support Vector Machines, Support Vector Regression, and Long Short-Term Memory (LSTM), to detect network anomalies in cellular traffic in real time. Sangaiah et al. [14] devised an automated procedure that integrates unsupervised learning with classification algorithms to detect and diagnose active failures in mobile networks by combining network-collected data with manual drive testing.

The data generated by 5G-MNCs exhibit inherent non-localized spatio-temporal dependencies and temporal state relationships between current and historical data matrices [15]. The inherent complexity of these spatio-temporal characteristics substantially increases anomaly detection difficulty for 5G-MNC data streams. Consequently, conventional statistical approaches and machine learning methods may be inadequate, requiring specialized computational architectures for 5G-MNC anomaly detection.

B. Deep Learning Approaches for Anomaly Detection

The effectiveness of deep learning-based methods in anomaly detection has been empirically validated across

multiple domains, establishing a robust theoretical foundation for further exploration and practical implementation [16], [17], [18], [19]. Particularly notable contributions include Thill et al. [20], who developed a temporal convolutional autoencoder (AE) for ECG signal anomaly detection. Li et al. [21] proposed a stacked variational autoencoder (VAE) integrated with GNNs to address time series anomaly detection challenges. Lunardi et al. [22] introduced an unsupervised deep anomaly detection system that employs convolutional autoencoding for network traffic profiling from initial packet sequences, incorporating adversarial training strategies to enhance detection robustness. Deng and Hooi [23] established a GNN-based predictive paradigm that identifies anomalies through systematic deviations between predicted and observed values. Liu et al. [24] employed a Temporal Convolutional Network (TCN)-based deep learning approach for channel prediction, with its computationally intensive tasks handled by Mobile Edge Computing (MEC) to reduce the response time. Huang et al. [25] devised HitAnomaly, a hierarchical Transformer-based model for log anomaly detection that effectively processes unseen log templates and anomalous parameter values.

Nam et al. [26] proposed the Dual-Time-Frequency (DualTF) model, which employs a nested sliding window mechanism: the outer and inner windows process temporal and frequency domain features, respectively; this synchronizes anomaly scoring across domains to resolve time-frequency granularity discrepancies. Lee et al. [27] introduced an Uncertainty-aware Dynamic Thresholding (UDT) method that dynamically adjusts detection thresholds by quantifying epistemic and aleatoric uncertainties in temporal data for time-series anomaly detection. Liu et al. [28] developed TopoGDN for fine-grained modeling of temporal and feature dimensions. The model utilizes multi-scale temporal convolution modules for feature extraction and incorporates graph structure learning with topological attention mechanisms. This combined approach effectively captures both inter-feature dependencies and high-order topological patterns. Dai and Chen [29] presented a Graph-Augmented Normalizing Flow (GANF) model that establishes causal dependencies among multivariate time series through Bayesian networks while employing normalizing flows for density estimation in anomaly detection. Wang et al. [30] devised COCA, a deep contrastive one-class framework combining contrastive learning with normality assumptions for time-series anomaly detection. Yang et al. [31] proposed DCdetector, featuring a dual-attention contrastive architecture with multi-scale patch design that generates multi-view temporal representations and enhances discriminative power through contrastive loss optimization. Meli [32] proposed a novel OUAD, using causal discovery to learn a normal causal graph of the system, and it evaluated the persistency of causal links during real-time acquisition of sensor data to promptly detect anomalies.

The above methods typically operate under the fundamental assumption that models cannot accurately reconstruct or predict unseen anomalous samples, thereby differentiating between normal and anomalous instances. However, their modeling approaches may inadvertently induce overfitting,

potentially enabling even anomalous data to achieve satisfactory reconstruction fidelity [33]. Furthermore, 5G network measurement data inherently has complex dynamic spatio-temporal characteristics and is susceptible to distribution shifts—making it imperative to develop novel methodologies tailored for 5G-MNC data.

C. Out-of-Distribution (OOD) Challenges With Temporal Shift

Inspired by OOD graph learning advancements, researchers have recently focused on temporal shift-induced OOD problems. For instance, CauSTG [34] introduces a causal framework that transfers both local and global spatio-temporal invariant relationships to OOD scenarios. CaST [35] employs a structural causal model (SCM) to interpret the data generation process in spatio-temporal graphs, and uses backdoor adjustment to decouple invariant features from temporal environments. STVE [36] encodes traffic data into two disentangled representations and incorporates contextual information from spatio-temporal environments—using them as self-supervised signals to enhance the generalizability of context-oriented representations and improve OOD performance.

However, current research mostly focuses on predicting temporal drift but has limited exploration of time series anomaly detection. Existing detection methodologies mainly address anomalies under identical training-testing distribution assumptions. However, experiments show significant performance degradation when facing data with diverse distribution shifts in real applications.

III. PROBLEM STATEMENT

This study formulates the anomaly detection problem in 5G Mobile Network Configuration (5G-MNC) as a spatio-temporal anomaly detection task. As illustrated in Fig. 2, the 5G-MNC system is formally represented as a weighted graph $G = (V, E, A)$, where the node set V comprises N base stations ($N = |V|$); the edge set E captures inter-base station connectivity relationships; the weighted adjacency matrix $A \in R_{N \times N}$ is constructed through geographical distance metrics, with each matrix element $A_{i,j}$ corresponding to the geographical distance between nodes v_i and v_j .

The 5G-MNC state at each timestep t is mathematically represented as a graph signal matrix $X_t \in R_{N \times C}$, where C denotes the feature dimensionality. To effectively capture temporal dependencies, we employ a sliding window of length T to construct spatio-temporal feature sequences $X_t = (X_{t-T+1}, X_{t-T+2}, \dots, X_t)$, as model inputs. The anomaly detection objective involves generating binary classification labels $y \in \{0, 1\}$ for given samples (the numbers 0 and 1 represent normal and anomalous, respectively).

Notably, this study adopts a semi-supervised learning paradigm: the training set X contains exclusively normal samples, while the testing set X' incorporates both normal and anomalous instances for model validation. This configuration effectively simulates practical engineering requirements in application scenarios where anomaly samples are scarce but require precise identification.

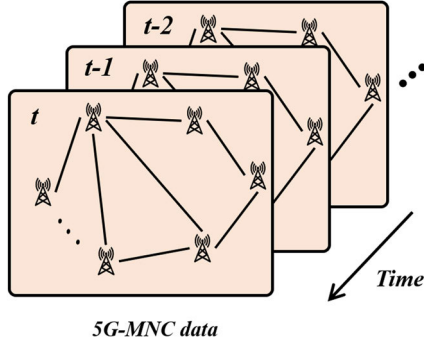


Fig. 2. Graph structure of 5G-MNC.

IV. PROPOSED APPROACH

This study introduces a deep spatio-temporal graph convolutional anomaly detection architecture, with its core architecture depicted in Fig.3. The methodology employs stacked Doubly Residual Spatio-temporal Graph Convolution (DRSTGC) blocks design. Each DRSTGC block contains two complementary graph convolutional modules: spatio-temporal graph convolution module (STGCM) synchronously extracts spatio-temporal correlations based on existing graph structures, and auxiliary spatio-temporal graph convolution module (ASTGCM) learns hidden spatio-temporal dependencies. To address distribution shifts in practical scenarios, a novel intervention module is integrated to mitigate confounding factor interference. The architecture ultimately performs anomaly detection through synergistic integration of discrimination and reconstruction modules. They effectively integrate information from both the original data domain and latent spatial representations. By integrating these modules with an adaptive scoring mechanism, the proposed model leverages their complementary strengths to achieve more robust anomaly detection. Subsequent subsections offer detailed descriptions of each individual component.

A. Temporal Embedding

The KPI metrics generated by 5G-MNC systems are significantly influenced by user behaviors and temporal patterns, consequently exhibiting seasonal variations that recur at periodic intervals (e.g., daily and weekly cycles).

To comprehensively model periodic dependencies in 5G-MNC base stations, we employ temporal embedding to encode each timestep into vector representations. For a given timestep, two one-hot encoded vectors are generated based on its hour-of-day and day-of-week attributes. These vectors are subsequently concatenated and projected through a fully connected layer to obtain temporal embeddings $m_t \in R^m$. For the t -th temporal segment containing historical timesteps, the corresponding temporal embedding matrix is formulated as $M_t = [m_{t-T+1}, m_{t-T+2}, \dots, m_t]^T \in R^{T \times D}$. The resultant temporal embeddings encapsulate critical timing information that is subsequently integrated into DRSTGC blocks for spatio-temporal modeling.

B. Doubly Residual Stacking Architecture

Inspired by the doubly residual architecture of N-BEATS [37], this study proposes a novel Doubly Residual Spatio-temporal Graph Convolution (DRSTGC) block to model implicit synchronous dependencies among base stations in 5G Mobile Network Clusters (5G-MNC). Our approach demonstrates significant architectural and functional innovations compared to N-BEATS:

Multidimensional Expansion: While N-BEATS addresses temporal dependencies in general time series through predictive sequence modeling, DRSTGC employs spatio-temporal graph convolutions to capture complex spatio-temporal correlations in input sequences. This extension enables effective modeling of both spatial topological relationships between base stations and temporal dynamics of signal propagation.

Functional Innovation: Diverging from N-BEATS's doubly residual architecture for time-series forecasting, our structure enhances representational capacity for anomaly detection tasks. As illustrated in Fig.4, the input $X_t^{(l)}$ to the l -th block generates two outputs: Reconstruction Output $\hat{X}_t^{(l)}$: Partial reconstruction results removing resolved explicit features. Hidden Representation $O_t^{(l)}$: Residual-connected features capturing unresolved latent patterns.

Progressive Feature Decoupling: The cascaded residual architecture enables hierarchical feature extraction. In the reconstruction path, previous modules progressively strip resolved explicit features $\hat{X}_t^{(l-1)}$, forcing subsequent modules to focus on unresolved information. The collective output of all blocks is formulated as

$$\hat{X}_t = \sum_l \hat{X}_t^{(l)} \quad (1)$$

$$O_t = \sum_l O_t^{(l)} \quad (2)$$

The final reconstruction $\hat{X}_t$ constitutes the aggregated summation of partial reconstructions from all blocks, achieving hierarchical decomposition while enabling interpretable output generation. Moreover, the reconstruction representation can uniquely identify this segment of temporal data as label information within the causal intervention module. Analogously, the ultimate hidden representation O_t , emerges as the cumulative integration of latent partial representations. Hidden representations can serve as core temporal information for causal intervention within the causal intervention module. This multi-scale aggregation mechanism facilitates multi-view representation aggregation and enables multi-scale latent pattern extraction, thereby enhancing anomaly detection sensitivity.

C. Doubly Residual Spatio-Temporal Graph Convolution Block

We propose the Doubly Residual Spatio-temporal Graph Convolution (DRSTGC) block to simultaneously capture temporal and spatial dependencies in 5G-MNC data. As depicted in Fig.5, the proposed architecture comprises two spatio-temporal graph convolution modules with doubly residual connections. The primary module systematically incorporates domain-specific prior knowledge, while the

auxiliary module subsequently processes residual information unresolved by its predecessor.

1) *Spatio-temporal Graph Convolution Module (STGCM)*: The STGCM is designed to synchronously extract spatio-temporal correlations through integrated Graph Convolutional Gated Recurrent Units (GCGRUs) [38] and fully connected layers. Given a graph $G = (V, E, A)$, the spectral graph convolution operation $*_G$ on input signal X can be formally defined as:

$$W_g *_G X = \sigma(LXW_g + b_g) \quad (3)$$

where $L = I - D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$ denotes the Laplacian matrix, I is the identity matrix, and D is the degree matrix. W_g and b_g are the learnable parameters. σ is the ReLU activation function. In the construction of GCGRU, the matrix multiplication in GRU is replaced with graph convolution operation. For convenience of presentation, omitting the superscript and the operation of GCGRU can be defined as

$$z_t = \sigma(W_z *_G [I_t; H_{t-1}] + b_z) \quad (4)$$

$$r_t = \sigma(W_r *_G [I_t; H_{t-1}] + b_r) \quad (5)$$

$$c_t = \tanh(W_c *_G [I_t; r_t \odot H_{t-1}] + b_c) \quad (6)$$

$$H_t = z_t \odot H_{t-1} + (1 - z_t) \odot c_t \quad (7)$$

where $I_t = X_t^* + m_t$ denotes the summation of input and temporal embedding, and $X_t^* \in \mathbb{R}^D$ represents the result of a fully connected transformation of X_t , aiming to have the same dimension as m_t . H_t denotes the output of GCGRU at time t . W_z, W_r, W_c, b_z, b_r and b_c are learnable parameters.

Then, two branches consisting of two fully connected layers are used to output partial reconstruction and partial hidden representation. For the STGCM in the l -th block, the outputs are defined as

$$O_t^{(l,1)} = \sigma(U_1^{l,1}H_t^{l,1} + b_1^{l,1}) \quad (8)$$

$$\hat{X}_t^{(l,1)} = \sigma(U_2^{l,1}H_t^{l,1} + b_2^{l,1}) \quad (9)$$

where $O_t^{(l,1)}$ and $\hat{X}_t^{(l,1)}$ are the partial hidden representation and the partial reconstruction of STGCM, and $U_1^{l,1}, U_2^{l,1}, b_1^{l,1}$ and $b_2^{l,1}$ are learnable parameters.

2) *Auxiliary Spatio-temporal Graph Convolution Module (ASTGCM)*: The ASTGCM serves as a complement to the STGCM, addressing information flows that cannot be resolved through prior knowledge derived from available graph structures. While the ASTGCM shares the same fundamental architecture as the STGCM, it distinguishes itself by employing an auxiliary adjacency matrix to autonomously learn unresolved hidden spatial dependencies. This auxiliary adjacency matrix is designed to complement the prior graph. Specifically, two randomly initialized node embeddings with learnable parameters are defined, and the auxiliary adjacency matrix is formulated as:

$$A_{aux} = \text{softmax}(\sigma(E_1 E_2^T)) \quad (10)$$

Here, E_1 and E_2 represent the source and target node embeddings, respectively, where their multiplicative product defines the spatial dependency weights. The ReLU activation

function is applied to eliminate weak connections, followed by a softmax operation to normalize the learned auxiliary adjacency matrix.

Subsequently, the self-learned auxiliary graph $G = (V, E, A_{aux})$ undergoes the same Graph Convolutional Gated Recurrent Unit (GCGRU) operations as the STGCM. The output of the GCGRU is then fed into two fully connected layers with ReLU activation functions to generate partial reconstructions and hidden representation outputs. The operations are formally defined as follows:

$$O_t^{(l,2)} = \sigma(U_1^{l,2}H_t^{l,2} + b_1^{l,2}) \quad (11)$$

$$\hat{X}_t^{(l,2)} = \sigma(U_2^{l,2}H_t^{l,2} + b_2^{l,2}) \quad (12)$$

where $O_t^{(l,2)}$ and $\hat{X}_t^{(l,2)}$ are the partial hidden representation and the partial reconstruction of ASTGCM.

3) *Block Output*: The outputs of the doubly residual spatio-temporal graph convolution block consist of two parts, one is the fusion of the partial hidden representations of $O_t^{(l,1)}$ and $O_t^{(l,2)}$, and the other is the summation of the partial reconstruction representations of $\hat{X}_t^{(l,1)}$ and $\hat{X}_t^{(l,2)}$. The outputs are defined as

$$O_t^{(l)} = \sigma(W^l[O_t^{(l,1)}; O_t^{(l,2)}] + b^l) \quad (13)$$

$$\hat{X}_t^{(l)} = \hat{X}_t^{(l,1)} + \hat{X}_t^{(l,2)} \quad (14)$$

where $O_t^{(l)}$ and $\hat{X}_t^{(l)}$ are the hidden representation and reconstruction outputs of the l -th block, respectively.

To address the interference of confounding factors in time series data under distribution shifts, this study proposes a causal intervention module. It mitigates the impact of distributional variations by introducing a PCIR regularization term. Therefore, $O_t^{(l)}$ and $\hat{X}_t^{(l)}$ are fed into the intervention module to compute the causal loss. Finally, the respective partial reconstructions $\hat{X}_t^{(l)}$ of each DRSTGC block are summed and fed into the reconstruction module to compute reconstruction scores. The partial hidden representations $O_t^{(l)}$ are aggregated and delivered to the discrimination module to calculate anomaly probabilities.

D. Intervention Module

1) *Invariant Representation Context Under Distribution Shifts*: Our formulation operates under the assumptions of an environmental set (denoted as E) and a sample space (denoted as X) of time series data. We consider a sliding time window sample X_t drawn from an unknown distribution p_n , which is primarily concentrated on a subset $N \subseteq X$ defining the normal class. Given $X_t \sim p_n$, we decompose X_t into a pair of random variables X_a and X_e such that $X = (X_a, X_e)$. Here, the environment E exclusively influences X_e . Conceptually, X_a represents the relevant features of time series X that determine its anomaly status, while X_e represents style features of time series X that determine its environmental status. While these style features remain unaffected by the anomaly status of the time series, they are influenced by environmental conditions.

Building on causal inference theory [11], we consider the process from X_t to O_t in Fig.4 as the encoder $f: X \rightarrow O$ that

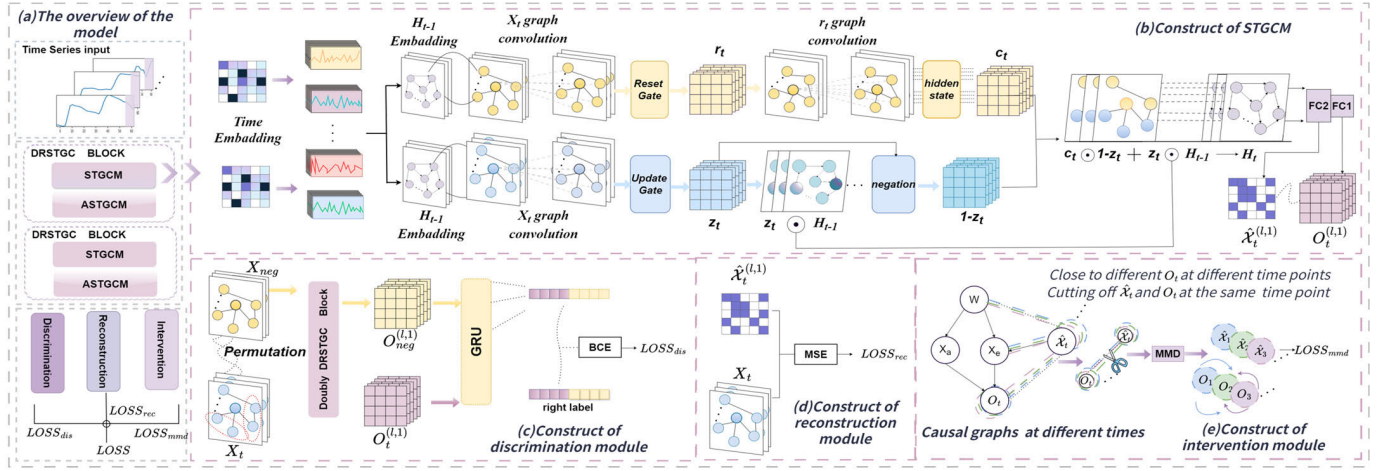


Fig. 3. Overall architecture of the model.

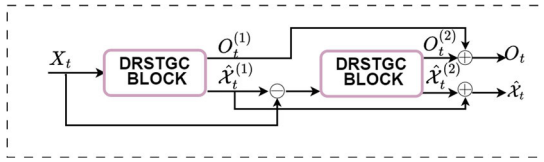


Fig. 4. Construct of doubly residual stacking architecture.

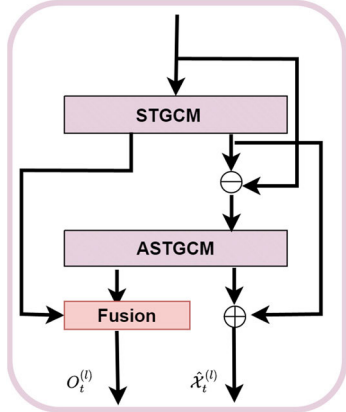


Fig. 5. Construct of Doubly Residual Spatio-temporal Graph Convolution block (DRSTGC block).

maps raw observations to a representation space O . Here, $O = f(X)$ constitutes the final representation. Domain invariance is formalized as follows:

Definition. We say O is an invariant representation of X under domain shift if

$$p^{do(E=e)}(O = \cdot) = p^{do(E=e')}(O = \cdot), \text{ for any } e, e' \in E. \quad (15)$$

The symbol do denotes an intervention operation. That is, the intervention $E = e$ is represented as $do(E = e)$, which differs from observing $E = e$ through existing data. p denotes probability. Invariant representations prevent anomaly detectors from misclassifying phenomena induced by domain shift. By definition, these representations O exhibit resistance to variations in E .

Therefore, this module aims to derive a causal loss function as a partial conditional invariance regularization (PCIR) term through causal intervention theory. This function trains the

Causal-DRSTGC model to obtain an invariant representation O of the original sample X . This enables more effective processing in the subsequent module for calculating more accurate anomaly scores. Below, we elaborate on the main workflow of the causal intervention module.

Firstly, we introduce a causal graph for anomaly detection based on a structural causal model (SCM), as illustrated in Fig. 6. We treat $\hat{X}$ as the environment E described above. As discussed in Section IV-B, $\hat{X}$ serves as the reconstruction representation of X , functioning to distinguish samples X and assign them to their corresponding environmental labels. Therefore, we further obtain:

We say O is an invariant representation of X under domain shift if

$$p^{do(\hat{X}=e)}(O = \cdot) = p^{do(\hat{X}=e')}(O = \cdot), \text{ for any } e, e' \in \hat{X}. \quad (16)$$

X_a and X_e denote relevant features and the style features, respectively. The hidden representation O represents the final representation of time series X . We introduce new random variable W that is a binary variable indicating whether a sample is normal ($W = 0$) or anomalous ($W = 1$).

It establishes causal relationships among five variables, where links from one variable to another indicate causation: cause $\rightarrow$ effect. We present the following interpretation of SCM:

$X_a \rightarrow O$: Since X_a is the primary feature and f is the mapping from X to O , any probability associated with X can be replaced by O . O also serves as the final output of the model, used to determine whether a sample is anomalous.

$X_a \leftarrow W \rightarrow X_e$: W fundamentally determines the sample attribute X , thus constituting a causal relationship, and consequently influences both X_a and X_e .

$\hat{X} \rightarrow X_e \rightarrow O$: The reconstruction representation $\hat{X}$ is treated as an environmental label at the causal level, uniquely determining the style features X_e within the sample. Subsequently, these style features exert a certain influence on O .

Theorem 1: Suppose that f learns an invariant representation, $O \perp \hat{X}$.

Proof. Since f learns an invariant representation and is measurable, O becomes X_a measurable. This implies that any

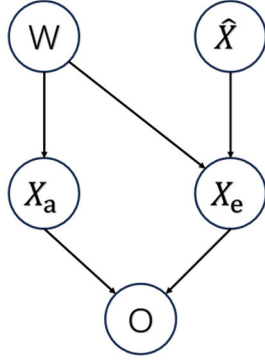


Fig. 6. Causal graph for anomaly detection.

probability involving events of O can be expressed in terms of probabilities involving X_a . It thus suffices to demonstrate $X_a \perp \hat{X}$. We prove this via d-separation. Note that there are two paths from X_a to $\hat{X}$. One via W and the other via O . The path via W is blocked because X_e is a collision, and neither X_e nor its descendant O are observed. The path via O is blocked because O is a collision with no descendants and is not observed. Hence, $X_a \perp \hat{X}$.

The fundamental challenge lies in ensuring statistical independence between O and $\hat{X}$. If this independence holds, we find that $p^{do(\hat{X}=e)}(O = \cdot) = p^{do(\hat{X}=e')}(O = \cdot)$, for any $e, e' \in \hat{X}$. By definition, O is an invariant representation of time series X .

Secondly, while counterfactual examples are generally inaccessible, enforcing appropriate conditional independence constraints can still induce counterfactual invariance. In practice, this condition may be formulated as $MMD(p(O = \cdot | \hat{X} = e), p(O = \cdot | \hat{X} = e')) = 0$, where MMD denotes the Maximum Mean Discrepancy. To capture complex high-dimensional differences, we utilize a cosine kernel, which is particularly effective in comparing the angular similarity between feature vectors.

$$L_{mmd} = \sum_{e, e' \in \hat{X}, e \neq e'} MMD \left(p(O_t = \cdot | W = \cdot, \hat{X}_t = e), p(O_t = \cdot | W = \cdot, \hat{X}_t = e') \right) \quad (17)$$

The causality loss function, as shown in Eq.(17), can serve as a PCIR term. Specifically, due to the temporal continuity, we randomly set the window value to 10. Additionally, there is a one-to-one correspondence between every element of $\hat{X}_t$ and O_t . For each value of $\hat{X}_t$ and O_t , we obtain the corresponding environmental label $e_i, i \in [1, 10]$, from reconstruction representation $\hat{X}_t$. And we obtain the corresponding hidden representation $o_i, i \in [1, 10]$, from hidden representation O_t . We then apply $MMD(o_i, o_j) = 0$, for any $e_i \neq e_j$, thereby obtaining the corresponding causal loss.

Furthermore, we compute the causal loss separately for normal samples ($W=0$) and anomalous samples ($W=1$), ultimately achieving a stronger decoupling between the reconstructed representation $\hat{X}_t$ and the hidden

representation O_t . This enhances the generalization capability of Causal-DRSTGC under distribution shifts. Moreover, the proposed model automatically generates environment labels through the reconstruction patterns decomposed by DRSTGC, eliminating the need for manual annotation and better suiting real-world application scenarios.

E. Discrimination Module

Within the discrimination module, the learned hidden representations are employed to distinguish anomalous data in the latent space. However, during 5G network operation and maintenance, generating sufficient anomalous samples for supervised training is usually infeasible— anomalies occur infrequently. To address this limitation, we implement a computationally efficient negative sampling strategy to generate pseudo-anomalous samples. Given a temporal segment X_t , we apply a permutation perturbation strategy to produce negative samples X'_t . Specifically, the permutation operation divides the signal into randomly sized subsequences and shuffles them, with each base station sequence undergoing independent permutation to deliberately disrupt inter-sequence correlations. Subsequently, random temporal jittering is introduced to the permuted segments.

The discrimination module processes hidden representations from either original inputs or negative-sampled data through a gated recurrent unit (GRU) layer, which sequentially computes hidden states across temporal intervals. The final hidden state is then passed to a fully connected layer with sigmoid activation to derive anomaly scores within $[0, 1]$ for each input segment, and to calculate the discrimination loss. The module is ultimately trained in an end-to-end manner using pseudo labels.

F. Model Training

Building upon the preceding analysis, we propose the utilization of MMD between two distributions as the driving mechanism for invariance induction. Additionally, we introduce Eq.(17) as a PCIR term specifically applied to anomaly detection in time series under distribution shifts.

The discrimination and reconstruction modules exhibit distinct yet complementary strengths, forming mutually reinforcing components. Specifically, the reconstruction module learns latent representations by reconstructing original observations, with anomaly detection achieved through reconstruction errors since unseen anomalies tend to induce substantial deviations. Nevertheless, the reconstruction-based module primarily focuses on sample reconstruction fidelity while ensuring distinctiveness between normal and anomalous sample representations. Consequently, this approach may enable accurate reconstruction of anomalous data.

The discrimination module is trained to identify anomaly in temporal evolution patterns, guaranteeing discriminative separability in the latent space. This effectively compensates for the limitations inherent in reconstruction-based methodologies.

Accordingly, the composite loss function is formulated as the weighted summation of three optimization objectives,

Algorithm 1 Training procedure of Causal-DRSTGC**Input:** Training data set $X = \{X_1, X_2, \dots, X_T\}$, Epoch E .**Output:** The learned model parameters Θ .

```

1: Initialize all parameters;
2: for  $e = 1, 2, \dots, E$  do
3:   for  $t = 1, 2, \dots, T$  do
4:      $O_t^{(1)}, \hat{X}_t^{(1)} \leftarrow \text{Block}_1(X_t, M_t, E_1, E_2)$ 
5:      $O_t^{(2)}, \hat{X}_t^{(2)} \leftarrow \text{Block}_2(X_t - \hat{X}_t^{(1)}, M_t, E_1, E_2)$ 
6:      $O_t = O_t^{(1)} + O_t^{(2)}$ 
7:      $\hat{X}_t = \hat{X}_t^{(1)} + \hat{X}_t^{(2)}$ 
8:     Generate negative sample  $X'_t$  and perform forward
       propagation to obtain the negative representation  $O'_t$  and
       the negative reconstruction representation  $\hat{X}'_t$ .
9:   // Calculate the total loss using Eq. (20)
10:   $L = L_{rec}(X_t, \hat{X}_t) + L_{dis}(O_t, O'_t) +$ 
     $L_{mmd}((\hat{X}_t, \hat{X}'_t), (O_t, O'_t))$ 
11:  Update model parameters using  $L$ ;
12: end for
13: end for
14: return The learned parameters.

```

Algorithm 2 Inference procedure of Causal-DRSTGC**Input:** Testing data set $X = \{X_1, X_2, \dots, X_T\}$, model parameter Θ , threshold τ .**Output:** The label y_t of X_t .

```

1: Load the trained model parameters  $\Theta$ ;
2: for  $t = 1, 2, \dots, T$  do
3:    $O_t^{(1)}, \hat{X}_t^{(1)} \leftarrow \text{Block}_1(X_t, M_t, E_1, E_2)$ 
4:    $O_t^{(2)}, \hat{X}_t^{(2)} \leftarrow \text{Block}_2(X_t - \hat{X}_t^{(1)}, M_t, E_1, E_2)$ 
5:    $O_t = O_t^{(1)} + O_t^{(2)}$ 
6:    $\hat{X}_t = \hat{X}_t^{(1)} + \hat{X}_t^{(2)}$ 
7:   Calculate the inference score  $s_t$  using Eq. (21);
8:   if  $s_t > \tau$  then
9:      $y_t = \text{"Anomalous"}$ 
10:  else
11:     $y_t = \text{"Normal"}$ 
12:  end if
13: end for

```

denoted as Eq.(20). Here, L_{rec} represents the reconstruction-based loss, defined as the mean squared error (MSE) between targets and reconstructed values. L_{dis} corresponds to the discrimination loss employing binary cross-entropy. L_{mmd} serves as the loss function for the intervention module, which incorporates the aforementioned regularization strategy. The hyperparameter α and λ balance the loss contributions of the three modules.

$$L_{dis} = - \sum_t \log(1 - D(O_t)) + \log(D(O'_t)) \quad (18)$$

$$L_{rec} = \frac{1}{T} \sum_{i=1}^T \| \mathbf{X}_{t+i} - \hat{\mathbf{X}}_{t+i} \|_2^2 \quad (19)$$

$$L = (1 - \alpha)L_{rec} + \alpha L_{dis} + \lambda L_{mmd} \quad (20)$$

In this architecture, D denotes the discriminator, where O_t represents the hidden representation of the input sample, and O'_t corresponds to the hidden representation generated by the negative sampling strategy. The training procedure for Causal-DRSTGC is outlined in Algorithm 1.

G. Model Inference

As mentioned in previous sections, the reconstruction and discrimination modules are jointly trained to optimize targets. As a result, both modules are encouraged to contribute to anomaly detection cooperatively. In order to do this, we create a scoring system based on reconstruction and discrimination that balances each of their contributions, increasing anomaly detection's overall efficacy. In particular, given an observation segment $(X_{t-T+1}, X_{t-T+2}, \dots, X_t)$, the reconstruction score is calculated from the deviation between the reconstructed values and the original observations, whereas the discrimination score, p_t is derived from the discriminator's output probability. The definition of the inference score s_t is

$$s_t = \frac{\sum_{i=1}^T \| \mathbf{X}_{t-T+i} - \hat{\mathbf{X}}_{t-T+i} \|_2^2 + \alpha \times (1 - p_t)}{1 + \alpha} \quad (21)$$

The hyperparameter α , which balances the contributions of the two components, will be thoroughly investigated in the subsequent sensitivity analysis section. An anomaly is identified when the inference score exceeds a threshold [39], where this threshold is determined through grid search on the validation set. The detailed inference procedure is presented in Algorithm 2.

V. EXPERIMENT

This section begins with an introduction to the dataset and experimental setup. Subsequently, the performance of Causal-DRSTGC is validated using real-world operational data from 5G-MNC systems. A case study is then conducted to analyze experimental outcomes, accompanied by ablation studies and sensitivity analysis. Finally, the model's performance is further validated on industrial datasets, and a complexity analysis is conducted.

A. Dataset Description

To evaluate the proposed method's effectiveness, we use a real-world 5G-MNC dataset. Collected by China Mobile Communications in Jinan, Shandong Province, it contains Key Performance Indicator (KPI) records from two 5G-MNCs. Cluster1 comprises 5 base stations, while Cluster2 consists of 14 base stations. The selected KPI for anomaly detection is the average number of Radio Resource Control (RRC) connections, with data aggregated at 15-minute intervals. The dataset spans from September 3, 2021, to January 3, 2022. Table I presents the partition lengths of the training and testing sets.

In practical applications, decommissioning and shielding represent two categories of anomalies in the operational maintenance of 5G networks, described as follows:

Decommissioning. Base station decommissioning refers to the phenomenon where base stations unexpectedly go offline during operation. It is caused by factors like severe weather and transmission failures, resulting in communication service interruption.

Shielding. As critical components of mobile communication systems, base stations play a pivotal role in maintaining communication quality. However, physical obstructions caused by billboards and outdoor buildings can degrade signal transmission. Therefore, base station shielding as an anomaly can affect the daily operation of 5G-MNC.

Fig. 7 illustrates two real-world anomaly scenarios involving 14 base stations in a 5G-MNC system, with each subplot depicting the average number of RRC connections. Fig. 7(a) demonstrates the decommissioning anomaly case, showing a simultaneous sharp decline in RRC connections across a group of base stations. As shown in Fig. 7(b) for the shielding anomaly case, a decrease in RRC connections at base stations B1, B3, B4, and B6 results in a corresponding surge at the neighboring base stations B2 and B5.

In 5G-MNC systems, anomalies are highly uncommon in practical scenarios, and existing anomaly datasets remain insufficient to validate the comparative performance of detection models. To address this, we implemented anomaly data imputation using the following methodology. Specifically, to simulate base station decommissioning anomalies, we stochastically reduced the number of users establishing connections with neighboring base stations at a specified rate of 0.6 over a given time interval. For base station shielding anomalies, we proportionally reduced the number of connected users at a target base station by a defined ratio of 0.6, then redistributed the reduced connections to its geographically adjacent base stations based on spatial proximity. Statistical metrics for both anomaly clusters are provided in Table I.

To more effectively simulate scenarios involving distribution shifts and render the distribution shifts in the testing sets more pronounced, while simultaneously enabling comprehensive demonstration of our proposed model's superior capabilities, we adopted a specific approach. Accordingly, we manually introduced distribution shifts into the testing sets of the aforementioned datasets by adjusting the magnitude of fluctuations to 0.5. Within the middle 70% of the temporal range, two continuous intervals were selected for each of the five feature columns, and variance manipulation was applied to approximately 10% of the data. This process retained the mean of the selected data segments while randomly altering the variance of individual data points.

Taking the testing set of Cluster1 as an example (total sample size: 3,542), two consecutive intervals were chosen for each of the five feature columns within the middle 70% of the time series. A total of 342 samples (approximately 9.66%) underwent variance perturbation. Features 1–4 each had 70 data points modified (1.98% per feature), while Feature 5 contained 62 modified points (1.75%). As exemplified in Fig. 8, the blue curve represents the original time series, and the orange curve illustrates the modified data distribution. This experimental design effectively validates the model's robustness under variance shifts conditions.

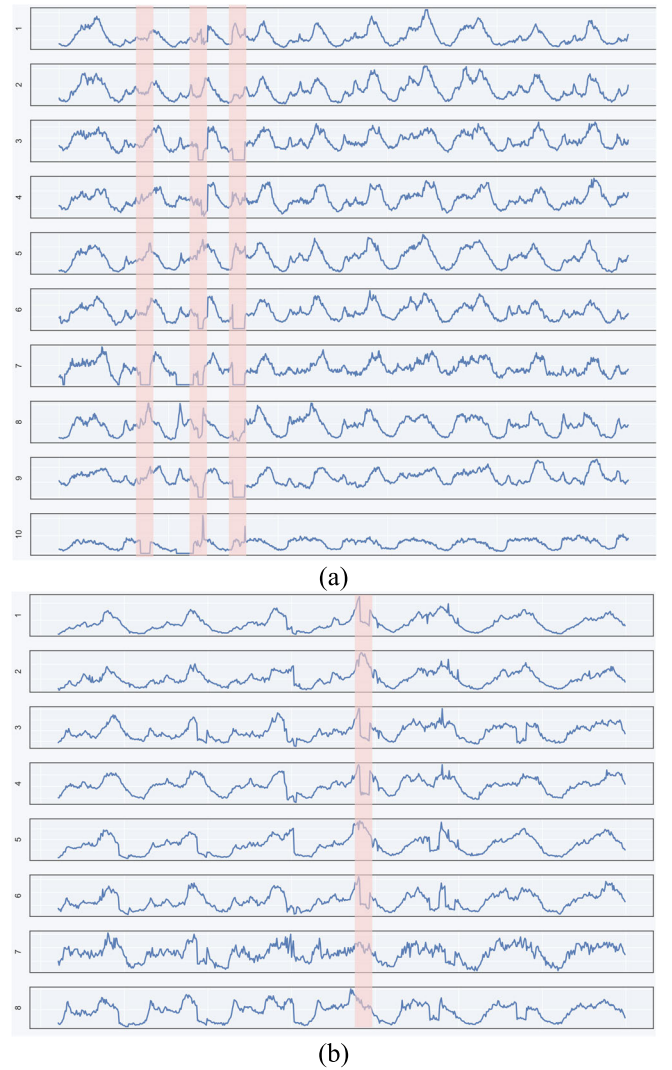


Fig. 7. Two real anomalies in the 5G-MNC data. (a) Decommissioning anomaly. (b) Shielding anomaly.

TABLE I
CLUSTER1 AND CLUSTER2 DATA STATISTICS

	Cluster1	Cluster2
Total length	11807	11807
Training length	8265	8265
Testing length	3542	3542
Number of series	5	14
Anomaly rate	3.14%	4.03%

B. Experimental Setup

Evaluation Metrics. To measure and evaluate method performance, four standard metrics were employed: precision (P), recall (R), F1-score (F1), and ROC-AUC (AUC) [40], [41]. In practical applications, anomalies typically manifest as continuous segments; thus, point-wise metrics are not prioritized. Following the evaluation strategy of [42], if any anomalous observation exists within a segment, the entire segment is labeled as the positive class.

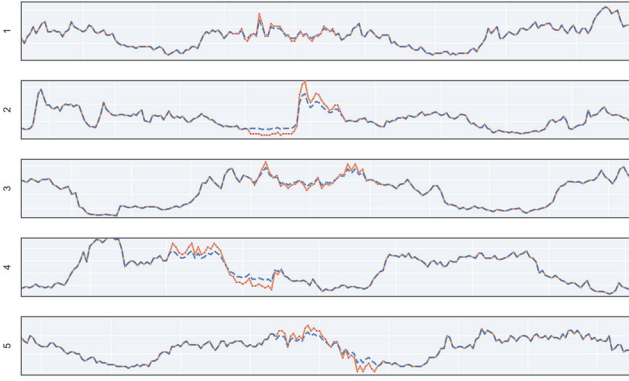


Fig. 8. Samples distribution shift for manual processing. The blue curve represents the original time series, and the orange curve illustrates the modified data distribution.

Baselines. The proposed method was compared against seven baselines: DCdetector (2023), GANF (2022), COCA (2023), UDT (2025), DUALTF (2024), TopoGDN (2024), and OUAD (2024). All comparisons adhered to the architectures recommended in their original publications. Since not all anomaly detection methods were designed for 5G-MNC scenarios, each model was evaluated across all feasible thresholds, and experimental results were reported based on the highest F1-score.

Data Preprocessing. Consistent with [43], Z-score normalization was first applied to the training and testing sets to enhance the robustness of the proposed method. Additionally, aligning with [44], the adjacency matrix for 5G-MNC was computed based on the geographical distances between base stations within the mobile network cluster. The weighted adjacency matrix A can be formulated as:

$$A_{ij} = \begin{cases} \exp(-\frac{\eta_{ij}^2}{\delta^2}), & i \neq j \text{ and } \exp(-\frac{\eta_{ij}^2}{\delta^2}) \geq \epsilon \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

where A_{ij} is the weight of the edge, and it is determined by η_{ij} , which is the geographical distance between base stations i and j . δ^2 and ϵ are the thresholds to control the distribution and sparsity of adjacency matrix A , and they are set to $1e-5$ and 0.3 in this work, respectively.

C. Experimental Results

In this subsection, we conducted experiments on the real-world 5G-MNC dataset to evaluate the effectiveness of Causal-DRSTGC against seven baseline methods. Table II presents the shielding anomaly detection performance comparison between Cluster1 and Cluster2, while Table III summarizes the decommissioning anomaly detection results. The best score achieved is highlighted in bold.

The proposed Causal-DRSTGC architecture comprehensively explores spatio-temporal patterns in original 5G-MNC data through specifically designed components. The STGCM and ASTGCM modules effectively leverage domain-specific knowledge to capture inherent synchronous dependencies within 5G-MNC systems. Discrimination and reconstruction

modules facilitate collaborative anomaly detection in both original data space and latent feature domains. Furthermore, the integration of causal graphs with intervention mechanisms effectively mitigates confounding effects under distribution shifts, significantly enhancing model robustness. Experimental results demonstrate that Causal-DRSTGC achieves state-of-the-art performance compared to existing methods.

Shielding anomalies differ fundamentally from decommissioning scenarios in 5G-MNC operations. While decommissioning involves simultaneous failures across multiple base stations within a region, shielding anomalies typically occur within individual base stations, where adjacent nodes compensate through dynamic traffic redistribution. This distinction results in stronger spatial dependencies during shielding events compared to decommissioning scenarios. Causal-DRSTGC effectively models evolutionary relationships between base stations and their neighbors through spatio-temporal correlation mining, with particular sensitivity to dependency alterations caused by shielding anomalies. This capability positions Causal-DRSTGC as a promising solution for 5G-MNC anomaly detection challenges.

To comprehensively evaluate the scalability and robustness of Causal-DRSTGC on large datasets with varying topologies, we expanded our data scope to clusters of different scales: 2, 4, 6, 8, 10, 12, 14, 16, and 19 nodes. Results demonstrate that Causal-DRSTGC maintains stably high performance across clusters of all scales, enabling reliable scaling to larger datasets and adaptation to diverse topologies. Experimental results are shown in Fig.9.

To rigorously test the model's adaptability to real-time environmental changes and dynamic topology shifts, we randomly altered the adjacency relationships among nodes four times within a 16-node cluster, simulating continuous network topology evolution. Experimental results demonstrate that our model continues to exhibit robust performance, effectively proving its stability and adaptability to dynamic 5G network environments. Experimental results are shown in Fig.10.

To validate the impact of long-range temporal and spatial dependencies across multiple nodes on anomaly detection, we further conducted the following experiments. Specifically, we set the time window size to 100 sample points. On the Cluster1 dataset, we removed one time window, three consecutive time windows, and three non-consecutive time windows respectively, and performed anomaly detection. The detection results are shown in Table IV, with the maximum values of the core metrics F1-score and AUC highlighted in bold. The results show that even a small amount of missing historical data has an impact on anomaly detection results, which proves that the model can extract effectively long-range dependencies from historical time-series data.

Subsequently, from a spatial perspective, on the Cluster1 dataset, we removed the adjacency edges of node 5, the adjacency edges between nodes 4 and 5, and the adjacency edges between nodes 3, 4, and 5, respectively, as illustrated in Fig.11. The detection results are shown in Table V, with the maximum values of the core metrics F1-score and AUC highlighted in bold. The findings indicate that the absence of link relationships from the original cluster reduces the

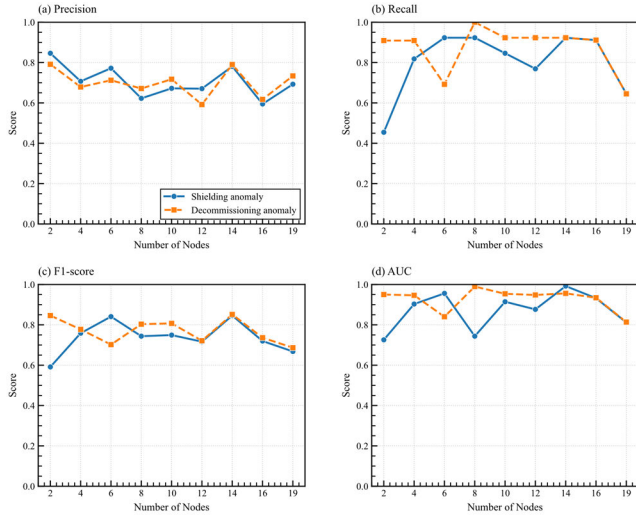


Fig. 9. Performance of clusters with different numbers of nodes. The blue curve represents shielding anomalies, and the orange curve represents decommissioning anomalies.

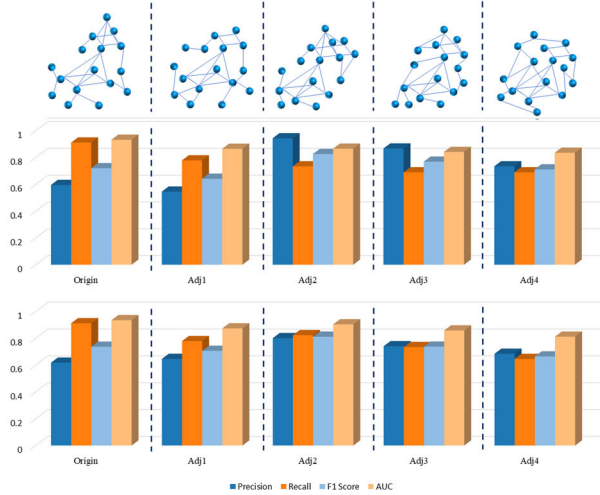


Fig. 10. Performance of different adjacency relationships in a cluster with 16 nodes, where the first row shows topological structures of different adjacency relationships among 16 nodes, the second row shows performance metrics for shielding anomalies and the third row shows performance metrics for decommissioning anomalies.

performance of anomaly detection to varying degrees, which demonstrates that the proposed model can capture spatial relationships between non-directly adjacent nodes for carrying out anomaly detection.

To comprehensively evaluate our model's adaptability beyond a single KPI, we conducted additional experiments using the signal strength (RSRP) metric on the original Cluster1 and Cluster2 datasets. As shown in Table II and Table III, the model still achieves competitive F1-scores and AUC values for both shielding and decommissioning anomalies.

D. Ablation Study

To investigate the effectiveness of distinct components in Causal-DRSTGC for anomaly detection, we employed the

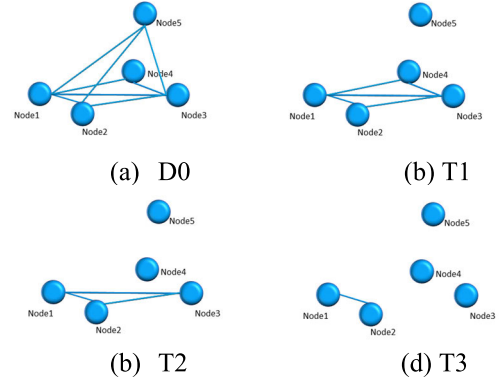


Fig. 11. Node relationship diagram after removing adjacent edges of specific nodes.

Cluster2 dataset for ablation studies in this section. Specifically, we disabled individual modules within Causal-DRSTGC for systematic evaluation. w/o STGCM and w/o ASTGCM denote the variants where the spatio-temporal graph convolution module (STGCM) and auxiliary spatio-temporal graph convolution module (ASTGCM) are deactivated in each block, respectively. The w/o Reconstruction variant performs anomaly detection exclusively through the discrimination mode, while the w/o Discrimination variant relies solely on the reconstruction mode. The w/o Intervention variant eliminates the causal intervention mechanism. As shown in Table VI, experimental results confirm the indispensability of all aforementioned modules in Causal-DRSTGC.

In the w/o ASTGCM configuration, only predefined topological information from 5G-MNC is utilized for representation learning, which restricts comprehensive exploration of 5G-MNC data characteristics. Similarly, disabling STGCM prevents full exploitation of prior knowledge, emphasizing the complementary roles of both modules. The absence of the reconstruction module confines anomaly detection to latent space analysis, neglecting informational cues from original data. Conversely, removal of the discrimination module prioritizes reconstruction fidelity while disregarding critical discriminative features that separate normal and anomalous samples. Eliminating the intervention module impairs the model's capacity to address distribution shifts, ultimately degrading detection performance.

To intuitively validate the effect of causal intervention module and the discrimination module, we provide a comparative visualization of the model's latent representations in Fig. 12. Specifically, we selected a continuous sequence with anomalies and visualized the model's output representations. We first reduced the dimensions to 2D space via UMAP, then generated scatterplots for visualization.

Fig. 12(a) illustrates the mapping of the proposed model, while Fig. 12(b) depicts the visualization without the intervention module. A stark contrast can be observed: the representations in Fig. 12(a) exhibit tighter, more structured clusters corresponding to normal and anomalous states. This suggests that the intervention module effectively disentangles the confounders, allowing the model to learn more invariant and robust representations that are irrelevant to distribution

shifts. In contrast, the features in Fig. 12(b) are more dispersed and overlapping. This dispersion indicates that the ablated model is susceptible to spurious correlations introduced by shifting data distributions. This visual evidence directly supports our quantitative results in Table II and Table III, explaining why the intervention module contributes to a significant accuracy improvement. That is, it guides the model to base its decisions on relevant features rather than on confounding factors.

Fig.12(c) presents the visualization without the discrimination module. It further elucidates that integrating the discrimination module into the hidden representation imposes direct constraints on the distribution of normal and anomalous samples in the latent space, thereby enhancing model performance.

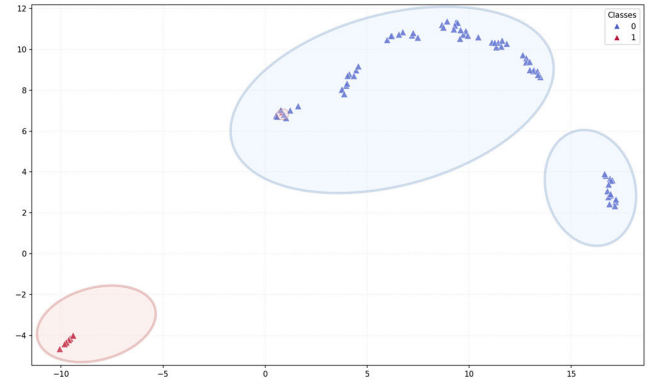
Collectively, architectural ablation of any module degrades detection efficacy.

E. Case Study

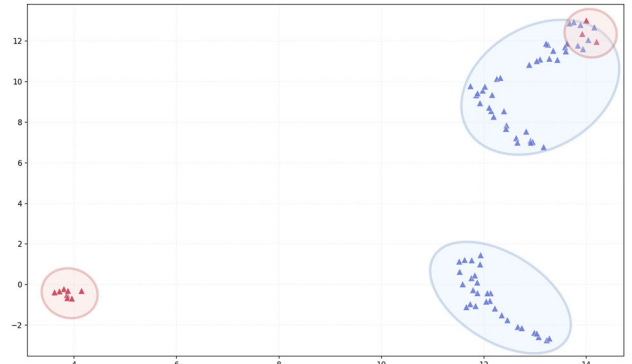
To visually demonstrate the anomaly detection performance of Causal-DRSTGC on 5G-MNC data, Fig.13 and Fig.14 presented two distinct visualization results. Specifically, we selected two anomalous 5G-MNC data segments from Cluster1 and Cluster2 datasets, respectively, then computed and displayed their anomaly scores. Fig.13 illustrates a shielding anomaly detection result on Cluster1, where node B3 exhibits a significant performance decline in the anomalous region, simultaneously inducing concurrent load increases in its associated nodes B1, B4, and B5. Fig.14 demonstrates a decommissioning anomaly detection result on Cluster2, where a group of neighboring base stations (B3, B8, B11, B12, and B13) show synchronized reductions in RCC connections within the anomalous region. Notably, in both cases, Causal-DRSTGC promptly generates conspicuous spike alerts.

Furthermore, in the architectural design of Causal-DRSTGC, the ASTGCM module employs an auxiliary self-learned adjacency matrix to capture unresolved hidden spatial dependencies, serving as complementary information to the predefined graph structure. To validate this mechanism, we conducted a case study visualizing the self-learned adjacency matrix. Specifically, we presented the geographical layout of base station B7 and its neighbors, along with the corresponding rows from both the predefined adjacency matrix and the self-learned matrix in Fig.15. Blue markers denote the top three neighbors specified in the predefined adjacency matrix for node B7, while brown markers indicate the top three neighbors identified in the self-learned matrix. The visualization reveals three additional base stations in the self-learned matrix that, despite lacking direct adjacency in the predefined structure, maintain relatively close geographical proximity and demonstrate measurable influence on B7's actual operational status. This evidence confirms that the self-learned matrix effectively supplements the predefined graph structure, enabling ASTGCM to uncover latent spatial relationships.

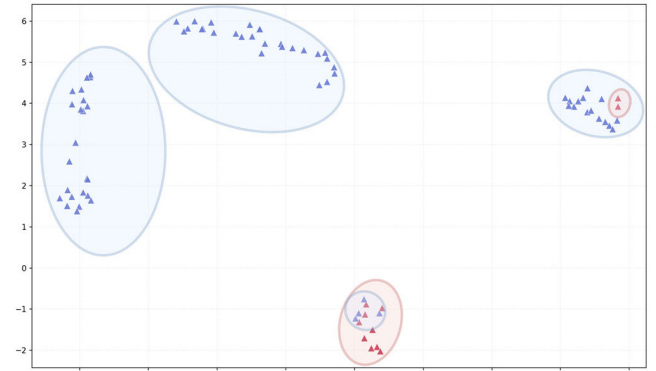
Given the multi-stage architecture with stacked DRSTGC blocks, there is a possibility of error propagation through the layers. We conducted a controlled experiment in which artificial errors were introduced after the first DRSTGC block. For



(a). Mapping of the proposed model in two dimensions



(b). Mapping of the proposed model lacking the intervention module in two dimensions



(c). Mapping of the proposed model lacking the discrimination module in two dimensions

Fig. 12. Mapping of different variants of the model in two dimensions (red dots are anomalous samples, blue dots are normal samples).

specific details, please refer to the Appendix A (Supplementary Material).

It is common for base stations to experience fault recovery scenarios in real applications. To validate the model's adaptability in such cases, we manually added more pronounced fault data to simulate the kind of transition processes and test the proposed model. For specific details, please refer to the Appendix E (Supplementary Material).

In addition, the impact of extreme weather is also a factor that base stations need to consider in practical scenarios. Hence, we simulated the data fluctuations and distribution

TABLE II
PERFORMANCE COMPARISON OF SHIELDING ANOMALY DETECTION FOR CLUSTER1 AND CLUSTER2

Method	Cluster1_shielding				Cluster2_shielding			
	P	R	F1	AUC	P	R	F1	AUC
GANF(2022)	9.52	66.67	16.67	86.84	50.50	50.50	50.50	97.98
DCdetector(2023)	11.06	33.33	16.60	54.23	16.75	50.50	25.10	50.75
COCA(2023)	68.18	20.69	31.75	--	77.97	21.71	33.97	--
DUALTF(2024)	3.42	100	6.61	48.06	4.04	100	7.76	48.04
TopoGDN(2024)	20.49	47.93	29.13	79.55	30.88	14.68	19.81	46.90
OUAD(2024)	100	8.50	15.67	--	100	10.00	18.18	--
UDT(2025)	8.47	90.91	15.49	78.24	8.26	38.46	13.61	52.16
Our model (RSRP)	65.13	81.82	72.53	90.13	91.67	69.23	78.89	84.48
Our model	80.66	100	89.29	99.57	78.10	92.30	84.61	99.21

TABLE III
PERFORMANCE COMPARISON OF DECOMMISSIONING ANOMALY DETECTION FOR CLUSTER1 AND CLUSTER2

Method	Cluster1_decommissioning				Cluster2_decommissioning			
	P	R	F1	AUC	P	R	F1	AUC
GANF(2022)	7.69	100	14.29	86.26	50.50	50.50	50.50	97.76
Dcdetector(2023)	20.28	66.67	31.10	59.46	6.04	16.67	8.87	44.89
COCA(2023)	54.66	14.87	23.38	--	77.36	21.20	33.28	--
DUALTF(2024)	11.75	38.84	18.04	57.18	9.11	26.57	13.57	41.95
TopoGDN(2024)	46.80	36.36	40.74	87.65	36.17	23.77	28.57	66.27
OUAD(2024)	100	0.85	1.68	--	100	15.00	26.09	--
UDT(2025)	59.14	45.45	51.40	80.56	54.32	30.77	39.29	62.60
Our model (RSRP)	84.62	81.82	83.19	90.65	70.06	76.92	73.33	87.77
Our model	90.16	90.90	90.53	95.27	79.04	92.30	85.16	95.63

changes caused by extreme weather conditions (such as heavy rain and blizzards) and further validate the performance of model. For specific details of experiments, please refer to the Appendix F (Supplementary Material).

F. Sensitivity Analysis

During model training, the hyperparameters α and λ balance the loss contributions of the reconstruction, discrimination, and intervention modules, which critically influence anomaly detection performance in 5G-MNC systems. To analyze the sensitivity of Causal-DRSTGC, extensive experiments were

conducted using the Cluster1 and Cluster2 datasets. Specifically, we evaluated the recall, precision, F1-score, and AUC for decommissioning anomalies and shielding anomalies under varying α values, with results visualized in Fig.16 and Fig.17.

For Cluster2, experimental results demonstrate that the model achieves optimal overall performance for shielding anomalies at $\alpha = 0.1$ and $\lambda = 0.1$, while the optimal configuration for decommissioning anomalies occurs at $\alpha = 0.3$ with $\lambda = 0.1$. The reconstruction module learns data distributions by minimizing the mean squared error between inputs and outputs. However, the scarcity of anomalous samples in detection tasks may render this module sensitive to

TABLE IV

PERFORMANCE COMPARISON OF TEMPORAL RELATIONSHIPS AFFECTED BY HISTORICAL TIME-SERIES. D0 INDICATES ORIGINAL DATA. D1 INDICATES DATA AFTER REMOVING ONE TIME WINDOW. D2 INDICATES DATA AFTER REMOVING THREE CONSECUTIVE TIME WINDOWS. D3 INDICATES DATA AFTER REMOVING THREE NON-CONSECUTIVE TIME WINDOWS

	Type	P	R	F1	AUC
D0	shielding	80.66	100	89.29	99.57
	decommissioning	90.16	90.90	90.53	95.27
D1	shielding	79.27	72.72	75.81	86.02
	decommissioning	84.61	90.90	87.64	95.16
D2	shielding	85.93	45.45	59.45	72.59
	decommissioning	20.88	54.64	30.20	73.60
D3	shielding	82.79	63.63	71.96	81.58
	decommissioning	91.66	54.54	68.39	77.18

TABLE V

PERFORMANCE COMPARISON OF SPATIAL RELATIONSHIPS AFFECTED BY NON-ADJACENT NODES. D0 INDICATES ORIGINAL DATA. T1 INDICATES THE DATA AFTER REMOVING THE ADJACENT EDGES OF NODE 5. T2 INDICATES THE DATA AFTER REMOVING THE ADJACENT EDGES OF NODES 4 AND 5. T3 INDICATES THE DATA AFTER REMOVING THE ADJACENT EDGES OF NODES 3, 4, AND 5

	Type	P	R	F1	AUC
D0	shielding	80.66	100	89.29	99.57
	decommissioning	90.16	90.90	90.53	95.27
T1	shielding	15.82	54.54	24.53	72.12
	decommissioning	5.07	90.90	9.60	65.27
T2	shielding	67.34	27.27	38.82	63.40
	decommissioning	9.72	72.72	17.15	74.38
T3	shielding	3.42	100	6.62	50.01
	decommissioning	3.42	100	6.62	50.01

noise. Concurrently, maintaining partial reconstruction constraints helps capture spatio-temporal dependencies among base stations, providing auxiliary information for the discrimination module. The discrimination module employs binary cross-entropy loss to distinguish normal samples from pseudo-anomalies. Higher weights prioritize discrimination capability enhancement, amplifying representational divergence between normal and anomalous samples to improve detection sensitivity for rare anomalies. The configuration $\lambda = 0.1$ balances robustness and flexibility, suppressing distribution shifts from environmental factors while preserving critical spatio-temporal patterns related to anomalies. Thus, for Cluster2, we select

TABLE VI

COMPARISON OF DIFFERENT VARIANTS

Method	P	R	F1	AUC
w/o ASTGCM	72.36	76.92	74.57	87.84
w/o STGCM	82.08	76.92	79.42	88.10
w/o Reconstruction	9.67	84.61	17.35	75.63
w/o Discrimination	70.39	61.53	65.67	80.23
w/o Intervention	75.00	92.30	82.75	95.50
Causal-DRSTGC	79.04	92.30	85.16	95.63

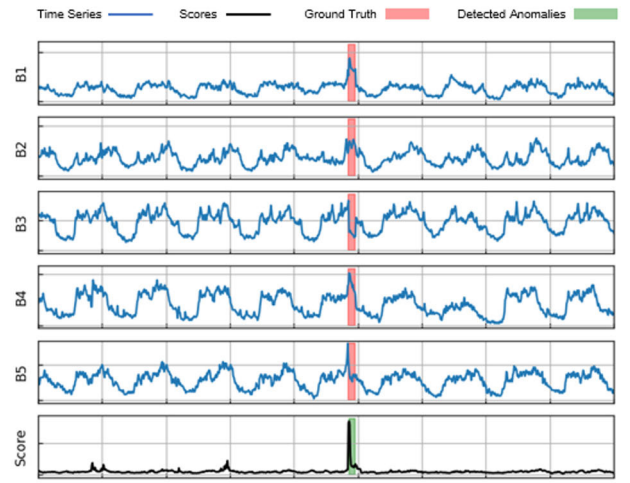


Fig. 13. Case of shielding anomaly detection result on Cluster1.

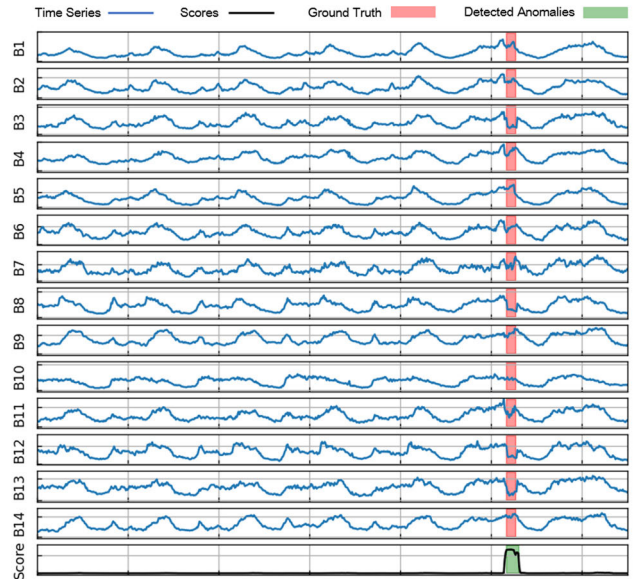


Fig. 14. Case of decommissioning anomaly detection result on Cluster2.

$\alpha = 0.1$ and $\lambda = 0.1$ for shielding anomalies and $\alpha = 0.3$ with $\lambda = 0.1$ for decommissioning anomalies.

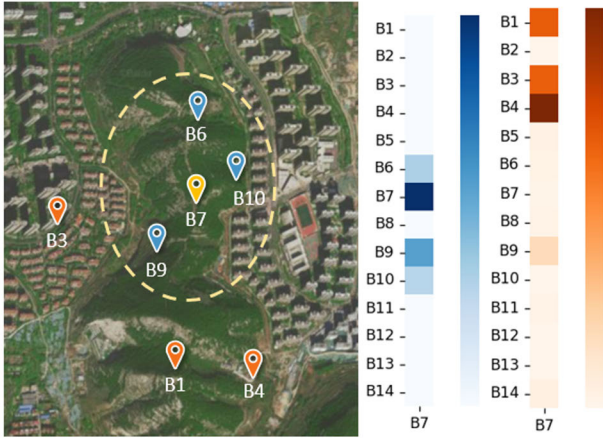


Fig. 15. Locations of base station B7 and its neighbors (each marker denotes a base station). The row of B7 in the adjacency matrix (blue) and the row of B7 in the learned matrix (brown).

For Cluster1, experimental results reveal that $\alpha = 0.5$ and $\lambda = 0.1$ yield optimal performance for both anomaly types. Compared with Cluster2, Cluster1 contains fewer base stations with lower feature dimensionality, exhibiting simpler spatio-temporal dependencies but more pronounced localized distribution disturbances. The higher α value (0.5) imposes stronger reconstruction constraints, effectively suppressing noise interference in low-dimensional feature space through enhanced input-output alignment while preserving essential spatio-temporal correlations. Furthermore, $\lambda = 0.1$ minimizes representation discrepancies across environments, mitigating amplified distribution shifts in low-dimensional features to ensure robustness under complex conditions. These experiments confirm that parameter selection must consider data characteristics: for 5G-MNC scenarios with smaller scale and lower feature dimensionality, we recommend $\alpha = 0.5$ and $\lambda = 0.1$.

We further conducted parameter sensitivity analysis with 2-node and 10-node clusters to investigate the adaptability of hyperparameters across different network scales. For specific details, please refer to the Appendix B (Supplementary Material).

Besides, in order to conduct a more comprehensive analysis of the model, we have further implemented sensitivity analysis from multiple perspectives including the influence of batch size on training efficiency and performance, the optimal stacking block number for DRSTGC, and the chosen of the negative sampling strategy. Due to space limitations, for specific details, please refer to Appendix C, D, and G (Supplementary Material).

G. Industrial Data Applications

1) Datasets: To further validate the generalizability of our proposed method, we conducted experiments using the following datasets. The PSM dataset [45], a public dataset obtained from eBay server machines, contains 25-dimensional metrics. The MSL dataset [46], collected by NASA, records sensor and actuator status data from the Mars rover. The

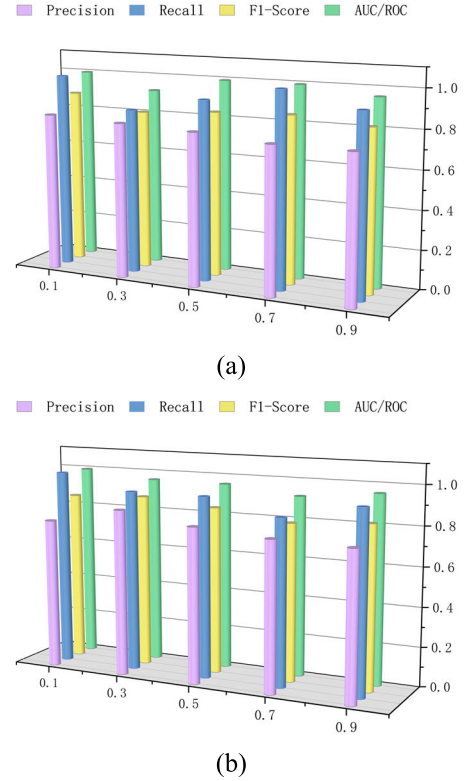


Fig. 16. Comparison of the performance of different parameters in Cluster2. (a) Shielding anomaly detection. (b) Decommissioning anomaly detection.

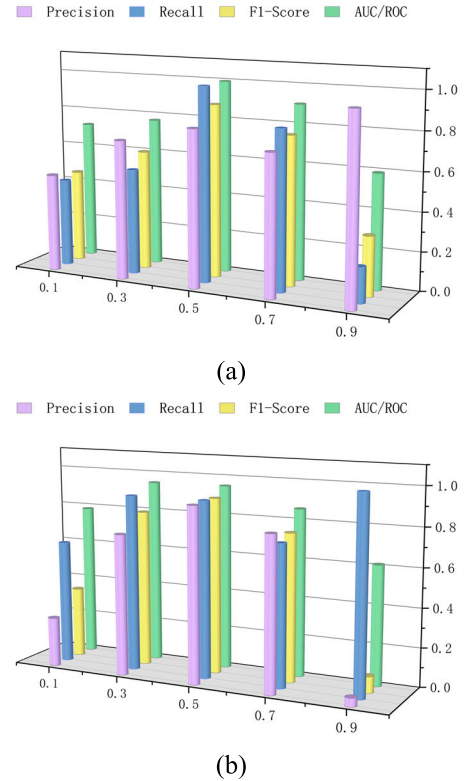


Fig. 17. Comparison of the performance of different parameters in Cluster1. (a) Shielding anomaly detection. (b) Decommissioning anomaly detection.

SMAP dataset [46], also collected by NASA, provides soil sample measurements and telemetry information from the

TABLE VII
PERFORMANCE COMPARISON OF PSM, MSL, AND SMAP DATASETS ON DIFFERENT BASELINES

Method	PSM				MSL				SMAP			
	P	R	F1	AUC	P	R	F1	AUC	P	R	F1	AUC
GANF(2022)	49.07	98.66	65.54	86.24	2.22	25.00	4.08	67.08	34.64	83.33	48.94	90.08
COCA(2023)	81.85	16.00	26.77	--	23.43	7.15	10.96	--	32.88	18.28	23.50	--
TopoGDN(2024)	38.59	55.57	45.55	60.28	12.85	85.60	22.34	55.09	14.17	85.29	24.30	47.35
Our model	78.47	89.07	83.43	89.84	67.40	87.81	76.26	91.40	89.41	53.40	66.87	76.24

TABLE VIII
PERFORMANCE COMPARISON OF THE PEPPER DATASET ON DIFFERENT BASELINES

Method	JointsControl				LedsControl				WheelsControl			
	P	R	F1	AUC	P	R	F1	AUC	P	R	F1	AUC
COCA(2023)	20.00	100	33.33	--	25.00	100	40.00	--	33.33	100	50.00	--
OUAD(2024)	69.35	100	81.90	--	67.79	100	80.81	--	52.50	100	68.85	--
DUALTF(2024)	39.35	100	56.48	50.74	37.73	100	54.79	46.58	25.22	100	40.28	41.61
TopoGDN(2024)	64.56	64.44	64.49	72.16	38.18	100	55.26	25.63	32.16	87.98	47.10	62.08
Our model	81.30	100	89.68	92.52	75.54	100	86.07	96.18	60.93	100	75.72	89.16

Mars rover. Compared with MSL, SMAP exhibits a higher incidence of point anomalies. Pepper is a humanoid robot developed by SoftBank Robotics for use in human–robot interaction and social assistance [47]. The dataset comprises 256 sensor readings. It records data for three distinct types of cyber-physical attacks: “JointsControl”, “LedsControl” and “WheelsControl”. The Pepper dataset is a well-known benchmark for anomaly detection in robotics, and is considered particularly challenging due to the high dimensionality of the data.

Similar to the 5G-MNC dataset, the testing sets were modified by introducing artificial distribution shifts through manual processing. Given the substantial data volume, our adjustment involved setting the fluctuation amplitude to 0.5 and applying variance perturbation to 30% of the data in the central 70% region.

2) Baselines: We selected the following advanced anomaly detection methods for comparison: TopoGDN(2024) [28], GANF(2022) [29], COCA(2023) [30], DUALTF(2024) [26], OUAD(2024) [32]. Our evaluation metrics included precision (P), recall (R), F1-score (F1), and ROC-AUC (AUC).

3) Performance Evaluation: We conducted comparative analyses of Causal-DRSTGC against baseline methods with results presented in Table VII. The experimental results demonstrate that Causal-DRSTGC achieves superior F1-scores across all three public datasets. Except for the SMAP dataset, Causal-DRSTGC achieves the highest AUC values on both PSM and MSL datasets, indicating its enhanced

capability to distinguish between normal and anomalous samples. Although GANF exhibits a marginally higher AUC than Causal-DRSTGC on the SMAP dataset, Causal-DRSTGC maintains significantly superior F1-score performance, suggesting a better balance between precision and recall.

We conducted experiments under three types of attacks on the Pepper dataset, with results shown in Table VIII. The findings indicate that our model also achieved the best performance. These experimental outcomes validate Causal-DRSTGC’s generalizability and effectiveness, particularly its significant advantages over existing baselines in balancing precision-recall tradeoffs, handling distribution shifts, and modeling complex spatio-temporal dependencies.

This demonstrates that the proposed model not only applies to 5G-MNC scenarios but also generalizes effectively to diverse time-series anomaly detection tasks, including server monitoring, aerospace sensor, and robot data, highlighting its practical utility.

H. Complexity Analysis

The time complexity of the DRSTGC block stacking architecture is $O(BT(N^2D + ND^2))$. The time complexity of the causal intervention module is $O(BT^2ND)$. Here, B represents the batch size, T represents the size of the time window, N represents the number of feature nodes, and D represents the feature dimension after data processing.

Besides, we calculated the actual running time and memory consumption of the proposed model on both Cluster1 and

TABLE IX
RUNNING TIME AND MEMORY CONSUMPTION OF
CLUSTER1 AND CLUSTER2

MNC	Train time	Train memory	Test Time/item	Test memory
Cluster1	1985s	13.86MB	0.07 $\pm$ 0.01s	8.35MB
Cluster2	1859s	37.04MB	0.07 $\pm$ 0.01s	23MB

Cluster2 listed in Table IX. The running time listed is the cumulative time for training dataset. We performed inference on the test dataset item by item to simulate real-world streaming scenarios, avoiding the use of any vectorization methods. One can observe that each item of data for inference requires approximately 0.07 seconds. Considering that the minimum time interval for mobile operators to collect each item of data is 15 minutes, the responsiveness of the model can meet the requirements of real-time applications. The experimental environment utilized an NVIDIA GeForce RTX 3090 GPU @ 128GB RAM configuration computer. These results demonstrate the capability of the proposed model for deployment in real-world industrial scenarios.

VI. CONCLUSION

In 5G Mobile Network Clusters (5G-MNCs), the proposed model effectively resolves complex spatio-temporal pattern recognition and distribution shifts robustness issues. The designed DRSTGC block achieves deep modeling of inter-base station synchronization dependencies through integration of predefined topological information with data-driven hidden dependency mining. By incorporating causal intervention mechanisms and partial conditional invariance regularization (PCIR), the architecture eliminates confounding effects induced by distribution shifts, significantly enhancing detection robustness in non-stationary environments. A joint optimization strategy combining discrimination and reconstruction modules leverages complementary information between original data and latent feature spaces, establishing a more reliable anomaly scoring mechanism.

Experimental results demonstrate that Causal-DRSTGC outperforms state-of-the-art methods on real-world China Mobile 5G-MNC datasets, particularly excelling in shielding and decommissioning anomaly scenarios with superior F1-score and AUC metrics. Ablation studies confirm the necessity of core components. Generalization experiments further reveal outstanding cross-domain performance on widely-used industrial benchmark datasets, highlighting strong universality and practical value.

Although this study demonstrated effectiveness, real-world 5G-MNC deployments still face challenges. First, the identification of anomaly types still relies on prior expert knowledge, particularly in dynamic and complex network environments, where autonomous anomaly categorization remains challenge. Second, the application background of mobile network clusters leads to the difficulty of root cause localization in real scenarios. Currently, there is a lack of systematic and automated

solutions to trace back to the fundamental causes, which hinders further improvements in overall operational efficiency. Hence, in future work, how to autonomously identify the emergence of new anomalies and locate the root causes of anomalies in 5G-MNCs are valuable research directions.

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